Corncob Redux - Earth Campaign

Lost Engines - Corncob Redux
Reviewed edition of the original game documentation.
Controls and changed rules describe Redux defaults.
Original credits, source tables and archival details are retained.

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Getting Started

Welcome to Corncob-3d!

Corncob-3d is an easy game to use.

If you just want to get in the cockpit and fly, then type <CR> at the main menu. Doing so will bring up the Training Missions menu. Just type <CR> again, and you will receive the pre-flight report, which gives information about the mission. Hit <CR> again to begin the mission.

Once in the plane, the blinking message reminds you that the F4 key gives inflight help.

The Story so Far

The date is August, 1949. This is not the 1949 that we remember, though.
In this 1949 WWII never happened (this is because among other factors, Adolf Hitler was killed at the age of 19 when hit in the head by a thrown
beer bottle), but all was not peace in this alternate age.

In 1938 aliens swept down out of the skies and began to set up structures. Although many attempts were made to communicate, the aliens did not
respond. It soon became obvious that these creatures simply were not interested in us, or in any other life on the planet. When human dwellings
or sites were in their way, the aliens destroyed them and continued their
seemingly random construction plans.

So in 1941 the world governments agreed to band together in an alliance
for humanity. No attacks were launched until November 1941. The plan was
a simultaneous all out blitz against the extraterrestrials in every part of
the globe.

Your mission is critical in this plan. You must destroy several alien structures, and in the process, many aliens themselves. If you do not succeed the entire human effort will fail.

Hardware Requirements

[Historical DOS documentation - retained for reference]

Hardware Requirements

  Corncob requires an IBM PC/AT-type, 12Mhz or faster, with a color VGA display, and 560k free RAM. A hard drive is recommended.

NOTE: If you are running Corncob from a high-density floppy disk: you must
copy COMMAND.COM from your DOS disk onto the root directory of this floppy
disk.

What to do if Corncob crashes your computer.

[Historical DOS documentation - retained for reference]

What to do if Corncob crashes your computer.

  Corncob has been tested on a variety of systems and has run reliably with no crashes for weeks of game play. However, Corncob 3D does do some
rather unorthodox things which may cause problems with certain TSR programs.

  If you encounter problems reboot with no TSR or menu programs loaded.

Redux Controls

These are the default bindings. F4 opens the current bindings during flight; Enter selects a binding to change it. Escape or P opens the pause menu, including display and control options. Delete clears a binding during capture; Escape cancels capture. Keyboard and mouse bindings can coexist: capturing a key changes the keyboard binding; capturing a mouse button changes the mouse binding. Delete clears both.

Flying
  Arrow keys or keypad 4/6 and 8/2: roll and pitch.
  Z / X or keypad 7 / 9: left / right rudder.
  + / - or keypad + / -: throttle up / down.
  Period or keypad 0: hold wheel brakes.
  Space or left mouse button: machine guns.
  Left Shift: heavy cannon. C or right mouse button: missiles.
  S: toggle sound effects. A: autopilot. F: flaps. I: rocket booster. B: drop bomb.
  E: step out behind the wing, or eject if airborne.
  R: rear view. L: cockpit. Num Lock: look mode.
  Hold Ctrl with arrow keys to look around.

On foot
  W / A / S / D: forward / left / backward / right.
  Mouse: look around. Space: jump immediately.
  Left Ctrl: drop through a supporting platform.
  Left mouse button: pistol. Q: open parachute.
  Semicolon: plant assassin charge. B: arm/detonate charges.
  V: call rescue. Caps Lock: remote control of your aircraft.

Shared controls
  T (hold): tenfold fast-forward while flying or walking.
  M: map. O: toggle radio messages. G: ground detail.
  P / Escape: pause menu. F1: boss screen.
  F2: intelligence. F3: current score. F4: control bindings.
  F6: cockpit / first person. F7: third-person view.
  Mousewheel: scroll menus, documentation and tower briefings.

Joystick buttons 1 and 2 fire guns and missiles during flight.
Modern joysticks do not need the DOS calibration controls.

Fast-forward advances the whole simulation, including enemies, projectiles and production. They do not sleep while you hold T.

Menus, documentation and the tower
  Arrow Up / Down: select or scroll; mousewheel also scrolls.
  Enter: accept. Escape: back/cancel. Backspace: erase text or go back.
  Home / End: first / last entry or beginning / end of a document.
  Page Up / Page Down: scroll documents by a page.
  Left / Right: adjust options; switch world groups in the world list.
  Insert: toggle detailed pilot records. Highlighted letters: shortcuts.
  Type a chapter number in the documentation contents, then Enter.
  In the tower, Enter selects a highlighted mission and closes the briefing; if airborne, this relocates your aircraft to that mission.
  Tower aliases: J/K or 2/8 scroll down/up; D/U move half a page; Space pages down; G goes to the start, Shift+G to the end; Q exits.
  Ctrl+F/N and Ctrl+B/P scroll; Ctrl+Y pages down, Ctrl+O pages up; Ctrl+T goes to the start, Ctrl+R to the end; Ctrl+H exits.

Redux Controls (continued)

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Map mode
  Arrows or keypad 4/6 and 8/2: move the map viewpoint.
  Page Up / keypad 9: descend (zoom in).
  Page Down / keypad 3: climb (zoom out). Hold S for slower zoom.
  Home / keypad 7: reset the map to your position.
  Hold Left Shift with arrows: rotate the map viewpoint.
  Escape, Enter, Space, or the configured Map key: return to flight.
  Map mode pauses the sortie; looking at a map does not advance enemies.

Mission builder (fixed editor keys, separate from flight bindings)
  In the world list, B opens the selected world in the builder.
  Arrows: turn the view; keypad 5/0: move forward/backward.
  Page Up / Down: lower/raise the viewpoint. 1..8: movement speed.
  Left Shift: show cursor or commit object; Left Ctrl: hide cursor.
  C: create an object. Space: select a nearby object. D: delete it.
  + / -: next/previous object type. ] / [: next/previous quality.
  Hold Left Alt with arrows and Page Up/Down: move cursor/object.
  Hold A with arrows: rotate selected object.
  F9: toggle special flag. F10: toggle objective-complete flag.
  Caps Lock: level the view. Alt+H: remember cursor height.
  H: move cursor to the remembered height.
  B, M or Escape: switch overhead/3D view. F5: save and return.
  Edits are exported under custom_worlds in the Redux save directory.

Other retained shortcuts
  Ctrl+C during flight: abort the sortie without awarding a score.
  Alt+I: enable invulnerability; in Other Worlds this toggles it.
  Escape stops a recorded demonstration without saving it.
  While a radio message is visible, O disables future messages;
  any other non-modifier key dismisses the message.
  F1's boss screen dismisses on any non-modifier key.

Mouse sensitivity and inverted look are adjustable in Options and
controls. Losing focus releases the mouse and pauses the game.
FOV, draw distance and ground detail do not affect simulation speed.
Ground-detail range changes how widely the dots are spread. A wider
range lets them shrink before the field ends. Increase the point count
as well if you want more nearby dots. The default range is eight times
the DOS field; choose 1 for the original extent.
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Keyboard Play

A great deal of effort was put into Corncob in order to make keyboard play work well. The keyboard is interrupt driven, and your key presses are timed to within 1/1000 th of a second. Also, Corncob has no problem detecting keystrokes even when 3 keys are held down at once.

Remember that to apply gentle tweaks to the control, you must press keys on and off rapidly, with the average time spent down being the desired roll/pitch rate. You have three displays on the front panel which show you what you are presently sending to the control stick. These meters are just to the right of the ALT light.

NOTE: if the keypad arrow keys suddenly appear to be not functioning properly, perhaps you have bumped the numlock key. See the 'inflight keys' section for the function of the numlock key.

Combat Missions: Theaters-of-Operation

To fly a mission a pilot first chooses in which theater of operations he wishes to fly. A theater of operations is a terrain filled with airports and areas of enemy concentrations. After opening a theater, select 'F' from the combat missions menu in order to fly a mission in that theater.

A pilot selects his mission by either flying there from the airstrip, or simply pressing F2 to bring up a list of the local missions. The pilot selects the mission by using the arrow keys to move the text until the start coords for the desired mission light up. Then by pressing the <enter> key, he/she will be teleported right to the action.

After you have completed your mission, you must fly back to your airstrip to get credit for your victory. If you can land the plane, your points will be much higher than if you quit during a 'flyby'.

Still, even if you cannot land the plane safely, quitting the flight via the pause menu must be done close to airstrip in order to avoid losing a plane. The definition of 'close' to your airstrip is when your airstrip's radar dish is blinking.

Most theaters of operation have 9 airbases from which the pilot can take off and land in. (However, the Earth campaign contains single-airstrip theaters.) Each airbase has intelligence information about the enemy forces in the vicinity. Intelligence info can be gotten from entering the control tower on foot, or by using your radio (F2 key) when in the air.
The information you receive will be from the last airport you were in.

One of the most important things about theaters is that they have a history capability. This means that anything a pilot has destroyed will stay destroyed forever. For example, say that yesterday a mission was flown in which the Devil's Aquarium was demolished. If that same theater is flown in today, the wreckage will still remain. This applies to friendly objects as well as enemy objects. Which brings us to the topic of plane conservation.

Planes are a precious resource which should not be wasted. Each time you crash a plane beyond repair, there is one less in your theater.

Even a heavily damaged plane can be repaired by landing and strolling over to the control tower on foot. If the pilot cannot land, then ending the mission with a low flyby will assume a safe landing with plane intact, and the next time the game is played the plane will be repaired.

If desired, some older model planes can be brought out of storage, but this should be used only when desperate. These planes have reduced performance specs, and are a brown color. Planes are brought out of storage when you select 'F' to fly at an airstrip where there are no regular blue planes available.

Combat Missions: Theaters-of-Operation (continued)

Of course, if a pilot runs out of planes in a theater, or would like a fresh start, there is always the option of reopening a fresh theater. A pilot can have up to 15 theaters open at one time, so he may fly one mission in DUSK theater, then fly a mission in WASTELND, and so on. Closing an undesirable theater does not affect the pilot's record. In fact even if a pilot is killed in action, he can be resurrected. The only cost of resurrection is the fact that the computer keeps track of how many resurrections you did, and places this number by your name in all records.

Each pilot has records kept both for his total career, and his performance in each theater in which he has flown.

Training Missions

If you don't want to fight in an actual theater -- you just want to hop in a plane and shoot at aliens -- then the Training Missions are for you. Use the menu to specify the number of enemy KLA sites, AAA batteries, and ground vehicle sites, and you are ready to engage in combat.

You will be placed in a universe which is empty except for your airstrip and the enemy field of operations. The basic setup of the enemy territory will always be the same. The fuel storage canisters (or Ticks, as pilots call them) are always the primary mission objective.

The default is that you are issued an invulnerable plane for training missions. If you wish the mission to add to your pilot's score, you may toggle off the invulnerable plane option.

Please note that the default training mission is much harder than the combat missions.

Friendly Objects

Airplanes
  Every airstrip has one or more airplanes that you may fly. To get into a
plane, simply walk up to it. To get out of a plane use the 'e' key. There
are three different plane types in the game.

  The blue-winged planes are the standard front-line ground attack planes.
The brown-winged planes are older planes which have been brought out of
storage in the case of an emergency. These planes have less engine power.

Control Tower
  Every airstrip has a control tower. This tower contains the radio
equipment and intelligence officers. If a pilot enters the door of the
control tower on foot, his plane will be repaired if close enough.

  If a control tower is destroyed, it may be impossible to get intelligence
info on future missions.

Radar Unit
  The radar tower watches for your safe return to home base. If you want
to have your plane on the next mission, you must make sure the radar tower
is flashing before ending the mission with the pause menu's End current
sortie option. The radar tower
begins flashing when you get to within a certain distance of the control
tower.

  Ending the mission with the radar tower flashing makes a big difference
in the computed pilot score.

Runways
  Runways are the ideal location for taking off and landing. Your plane
can do both on the grass also, but the risk of crashlanding is much higher
when not landing on the runway. Taxiing on the grass bounces the plane
about, and can cause a crashlanding even at low speeds.

Rescue Van
  When a pilot is downed behind enemy lines, he can be rescued by the rescue
van. He must first eject from the plane, and then use his handheld portable
radio to call for assistance. (Press the 'v' key.) A rescue van will be
dispatched and will appear shortly. The van will approach you and stop when
within a few steps of the pilot. When the pilot enters the van the mission
is over, and the pilot is considered home safe.

  The rescue van is not indestructible. The van will be destroyed if it

Friendly Objects (continued)

collides with another object on its way to pick you up. It is up to the pilot to maneuver himself so that there are no obstructions between the approaching van and himself.

Enemy Defenses

Force Fields
  Force Fields are a passive defense system used to protect resources by shielding them from firepower, and preventing our aircraft from flying through. Some forcefields shimmer, some are solid colors. Some fields appear as a rectangular frame which may or may not vary in size with the passage of time.

  The solid and shimmering force fields require the most power, and therefore only click on when needed. When a bullet or missile, or a plane itself comes near a solid forcefield, it will pop into existence.
Almost all bullets and missiles are stopped by the forcefields, as well as bombs. Do not use missiles when your plane is too close to field, as they will explode in your face when launched. A forcefield will cause damage to a plane at a certain rate during the time in which the airplane is flying too close. If a forcefield must be flown through,
then to minimize damage, fly through perpendicular to the surface, and the higher the airspeed, the better.

  In almost all cases, forcefields are powered by a local generator orb. Therefore when you take out the orb, all the local ff's will drop.

  Note that the enemy's fire can be stopped by forcefields also.

Barrage Balloons
  Barrage balloons are huge membranes filled with a lighter-than-air gas.
They float up and down randomly. Their purpose is simply to be in the way.
They are organic and self-regenerating, so although they stop all our firepower, they themselves are indestructible.

Bee Swarms
  Bee swarms are very similar to the barrage balloons discussed above. They
float up and down, but they are much harder to see, and are more difficult to
determine their exact position due to their indistinct appearance.

AAA Batteries
  These are one of the most threatening of the enemy's active defenses. AAA
sites track your plane and send up volleys of explosive shells which burst in
the air around you. You do not have to be hit directly to take damage from
an AA shell. The shells are designed to explode into fragments that tear
through your plane, even if you are hundreds of feet from the explosion.

  The AAA batteries vary wildly from site to site. We have found that there
is a correlation between muzzle color and deadliness. Usually brown AAAs
are the least threatening, while white AAAs are devastating.

Enemy Defenses (continued)

Note that enemy AAA shells can destroy enemy KLA's.

KLA's Deathballs

These are enemy missiles that lock on to your plane, and attempt to explode when close enough. They are launched either from missile sites or from saucers. These missiles have varying abilities which you can determine from their color.

When a KLA becomes active, it will show up as a green dot on your radar screen. When you see one visually you will notice that when it is pointed at you it begins to flash. At this point the KLA has a lock on you. Most KLAs will accelerate forward when they get a lock.

Note that deathballs always head directly towards you, and do not lead your plane, like AAA batteries do.

Generator Orbs

Generator orbs produce power for force fields and act as a central structure to each concentration of enemy activity. The process used by the aliens to generate power has a strange disorienting effect on humans when close to the orb. Usually this effect decreases with distance, but under certain circumstances can become independent of position.

When attempting to destroy it, aim for the center of the flashing ball.
It takes maybe ten missiles or two bombs and a few missiles to destroy one.

When an orb is destroyed all the force fields in the immediate vicinity
drop. This makes them of extreme strategic importance on some missions. In some situations it is wise NOT to destroy the orbs, since sometimes the alien's force fields can be used to our advantage.

There is a special kind of orb which has an attractive tractor beam that
exerts a pulling force on your plane.

Some orbs have been designed to be invincible to normal weapons, and the
only effective way to take them out is via suiciding the plane into them.
Make sure you bail out before contact, or else you will be killed in the ensuing explosion. If you must bail a good way away from the target, hit
caps lock to control your plane remotely.

TRFRU

Tetrahedral Radio Frequency receiving units, or TRFRU, are antenna systems used for spying on all human military communications. They look like small black tetrahedrons spinning in the air.

Mortars

Mortars are simple ground devices that spew incendiaries when any human
is near. They are used to deny low altitude flying.

Mind Benders

Enemy Defenses (continued)

Mind benders are small tofu-like structures that have a hallucinatory effect on the human nervous system. These devices blaze with flashing colors, and spin slowly about the ball in their centers. They are extremely tough, and can only be taken out with a bomb. In many missions they are unreachable from the air. In this case a volunteer must land his plane in enemy territory and proceed on foot to plant an assassin bomb on the device, then blow it up remotely from a distance.

Aliens

Aliens are very light and like to be in constant motion. They rarely are seen on open ground and must be approached on foot most of the time. The handgun is the weapon of choice for assassinating aliens. Upon death chemical reactions occur that may cause the body to float up into the sky.

If an alien touches you while you are not in your plane, you are captured.

Ground Transports

These ground vehicles look like three wheeled radio controlled cars. They are no threat to the airborne pilot; however it is a different story for the pilot on foot. These shopping carts (unofficial name) can capture an ambulatory pilot.

There are two ways to avoid capture while on the ground. One is to use your handgun to destroy approaching carts. The other method is to lead them into some wreckage or other structure which will cause the cart to turn onto its side and die. Ground vehicles are notoriously stupid.

When a shopping cart comes near, a human on foot feels a force inexorably pulling him towards the cart.

Flying Saucers

Saucers are the primary vehicle for alien airborne transportation. They vary widely in self-defense capability. Some possess AAA; some have missile launchers (KLA) aboard. All but the dumbest have the ability to execute evasive maneuvers. All saucers have the ability to make bombing runs against airfields. The color of the top ball on the saucer indicates its intent.

Saucers emanate from saucer sites. These sites look like large square cookie tins with an 'x' in the bottom.

Portals

The aliens have developed a means of instantaneous travel. Portals are

Enemy Defenses (continued)

small force fields which look like rectangles with the corners clipped. There are a number of different types of portals. Some are used by saucers to appear directly over the home airstrip. These can be used by the clever pilot as a means of getting home when his plane is not serviceable enough to fly back.

Other kinds take you to a random place in the theater, depending on whatever the aliens used it for last. Others are placed as traps that result in a destroyed plane, or worse, instant death.

How to Take Off, Land, and Fly Level

The key to success in Corncob is to become a good flyer. Only when a pilot learns the abilities and limitations of his plane can he make the complex maneuvers required to make it to the rank of General.

Takeoffs

Pulling your plane up into the air is the easiest part of flying. Merely put the throttle to the maximum, and pull up on the stick when you reach over 150 mph.

However, there are some details to be aware of. If your plane is not lined up on the runway, you will have to maneuver on the ground. The joystick controls the tilt of the plane, it cannot turn the plane. Therefore, use the rudder controls to steer while on the ground.

To bring the plane to a stop you must use the brakes. These are the Period key, the '.' key, or the '0' key on the keypad. The brakes must be used to bring the plane to a complete stop. (The plane must be brought to a complete stop to qualify as a landing.)

Once you are lined up, bring the throttle up to max. When you reach 100 mph the tail of your plane will rise off the ground. Keep the plane on the runway by tapping the rudder keys if needed. When you surpass 150 mph, pull back gently on the stick. If you pull back too hard the tail will swing down so quickly that it will slam back down onto the runway, thus kicking down your nose due to the bounce. This can cause a crash, so be careful.

Level Flight

When in the air the most important thing is to avoid crashing into the ground. To achieve level flight you must keep your gunsight above the horizon. How high you must keep your nose depends on your airspeed. Near the stall speed of 100 mph, the nose must be kept much higher above the horizon than when you are flying at 350 mph.

There is an altimeter and a climb rate meter to help you fly level. A warning light will flash if your altitude drops below 150 feet.

When climbing, keep an eye on your airspeed. If this drops below 100 mph, then you may not have enough airspeed to fly. This makes a stall possible.

Stalls and Advanced Aerodynamics

A stall occurs when the air traveling over the wings becomes turbulent. This results in a dramatic loss of lift, and unless corrected, the plane will invariably fall to the ground. Stalling can be brought about by

How to Take Off, Land, and Fly Level (continued)

letting your airspeed drop below stall speed, or it can also occur as a result of an imprudent aerobatic maneuver.

In a computer sim, the joystick tension does not reflect the flight situation. Therefore, Corncob has in it an automatic stall preventer that limits the stick travel to the maximum attainable without causing a stall.
This does not mean that it is impossible to stall. The stall preventer simply limits the control stick from causing a stall which would be the result of pulling back too hard.

The most common cause of a stall is simply that the airspeed has dropped too low to sustain flight. Another common cause is excessive sideslipping.

Also, a damaged plane is much more likely to stall. Wing damage will lower the lift of the wings, and control damage can make the stick jitter beyond what the automatic stall preventer can handle. Of course engine damage will lower airspeed, and rudder damage will increase level flight sideslip.

How to get out of a stall

To get out of stall means getting your forward airspeed up. Since your direction of motion is invariably down when stalled, it is best to get the nose pointed down. Of course your throttle should be place to maximum also.

Note that when the stall is so bad that your airspeed is nil, then you have lost the ability to control the plane with the control surfaces. In fact, it is even possible for your plane to fly backwards with the controls operating in a reversed fashion. Even backwards airspeed is better than zero. If you are falling backwards, pull the stick back to push the nose down. Once the nose is down, hold her there until you get some forward airspeed.

The emergency rocket booster can also be useful in getting out of a stall.

Landing

This is one of the hardest parts of learning to fly.

The biggest problem for beginners is to get lined up properly. One point to remember is that you must not fly toward the middle of the runway. Fly towards the close end of the runway, or a point on an imaginary line running the length of the runway and beyond. Once you are over that imaginary

How to Take Off, Land, and Fly Level (continued)

line,
then turn your nose to point towards the end of the runway. Getting lined
up is especially difficult in Corncob, due to fact that the plane may be slipping sideways.

  When you get close enough to the runway so that it no longer looks like a
line, but instead looks like a black rectangle, engage the flaps, and drop
your throttle to zero. (But don't let your airspeed drop below 100 mph.) At
this point you should be no higher than 1000 ft.

  Your plane has automatic landing gear that pops down as required.

  Now let the plane drop, shedding that altitude before you reach the end
of the runway. When you get to 100 or 200 feet, level the plane out for the
final drop to the runway.

  If you are not perfectly lined up, but are low and over the runway, then
drop and steer with the rudder once your wheels touch. The gear of your
plane is quite tough, and on the runway surface you can have your rate of
climb meter almost pegged negative and still survive. Landing on the
grass can be done, but requires a much more sensitive touch down.

  It is a good idea to be holding down the brake key throughout the whole
landing process. The Period key is situated conveniently for the
left hand to operate the brakes and rudder controls. For those that use
the left hand on the joystick, use the keypad rudder and brake controls.

  Once you have solidly touched down you may taxi about using your rudder.
Be careful to keep your speed down while taxiing (especially if your flaps
are down.) because it may cause you to lift into the air inadvertently, and
then cause a crashlanding. Please remember than on the ground, the stick
is useless, and you must steer with the rudder keys. Pulling the stick over
on the ground may cause your plane to flip.

Crashes
  There are two different types of crashes. One is a fatal wreck, and the
other type is a crashlanding. A crashlanding is where your plane is brought
down too hard, and the landing gear fails. The plane then spins to a stop
on the ground. When it stops, your plane is dead and will not fly again
until repaired. If you were close to your airstrip, the repair crew can fix

How to Take Off, Land, and Fly Level (continued)

it. (Do this by entering the control tower through the door on foot.)

  The other type of crash is the fatal crash. This happens if you slam into
the ground too hard, or if your plane is so tilted during landing that your
wing tip hits.

Midair collisions
  There are many types of objects which will destroy your plane if you collide
with them. When this happens your plane bounces, and the damage indicators
immediately max out. The screen border turns dark red, and the eject light
begins to flash. At this point you should bail out, since it is likely that
you will die if you try to land the plane. Don't bother trying to make it
back to base. (Although landing such a plane has been done on occasion,
it most often proves to be a fatal mistake.)

  Some of the objects which will destroy your plane in a collision are:
Saucers, fuel dumps, TRFRU, generator orbs.

How to Turn Your Plane

Please note that Corncob's advanced flight model makes turning different than on many commercial flight games which have simpler models. With a little practice however, you will find the plane is just as maneuverable.

To turn your plane, simply move the stick left or right. The plane will begin to bank. Don't be too shy, give the plane a nice 45 degree bank. Now that wing's lift is no longer pulling straight up, you will lose some lift, and your nose may drop. Pull back on the stick to keep the nose above the horizon. You will notice that as you pull back on the stick, your nose will be pulled into the turn.

To make a typical combat turn, place the plane on its side, so that the horizon forms a vertical line. Then pull all the way back on the stick, and hold it there until you have the compass reading you desire. Then simply flatten your plane back to the level. Be careful not to let the nose drop during the turn. You can always bring it up by unbanking the plane a little, and the nose will lift since you are holding the control stick back. This method of turning is actually easier than a slow coordinated turn. The price for the fast turn is airspeed.

The fastest turns are done with the flaps engaged, since this greatly increases the lift from the wings. Be careful not to let your airspeed drop too much. Flaps increase drag also.

Note that the amount of time required for the turn depends very much upon your airspeed. (Just like in an automobile, when moving fast you cannot make tight turns.)

Running Around - Welcome the Assassin

Part of your job as a pilot involves special forces missions. You may have to land in enemy territory and continue on foot. You carry a pistol with twenty rounds and one assassin charge; boarding a plane resupplies you, including a crashed plane.

Move with WASD and look with the mouse. Space jumps immediately; you can steer in the air. Q opens the parachute, even during a jump, and Left Ctrl
lets you drop through a platform. Horizontal force fields, fuel dumps, platforms and control towers may support your weight. Watch your landing:
a long fall can still kill you.

Plant a charge with Semicolon and move to safety. B arms the charge and lets it fall; contact then detonates it. A charge can be planted in midair.
Mindbenders are vulnerable only to bombs, so some missions require you to
plant a charge inside one and retreat before arming it.

Shopping carts and aliens can capture you. Use your pistol, jump to safety, or lure a cart into wreckage. V calls a rescue van. Caps Lock switches remote aircraft control; it does not move you back into the cockpit. Walk near a parked plane to board it automatically.

Scoring and Medals/promotion

Scoring
  Your score for a mission is determined by a formula outlined below. Your
raw score which depends on how much stuff you killed is multiplied by 3
factors. These three factors depend on what happened during your
mission.
For example say your raw score for a mission was 1000. Further suppose
that
you returned from your mission, and landed your plane safely at an
airstrip.
Then your final mission score would be:

5 * 5 * 1 * 1000 = 25000

   But say you ejected in combat, and had to be rescued by the rescue
van.

4 * 1 * 1/2 * 1000 = 2000

  Here you can see that bringing your plane home increased your score by
10 times.

  Now let's say you brought your plane back, but didn't have the guts to
try and land it.

5 * 1 * 1 * 1000 = 5000

  So you can see it is worth it to learn how to land the plane. It may
be
impossible to get a Stellar Cluster without landing your plane.

  If you make multiple flights during your mission, the multiplier
reflects
the worst thing you did. For example, if you crashland a plane, but
then
take another into the air, (without quitting the mission) and do a great
job, and even land beautifully, your multiplier will reflect the fact
that
you crashlanded. The fact that you also landed will not affect your
score.

  Therefore you can see that flying against multiply mission objectives
may
raise up your score, but you are jeopardizing the points you have
already
earned by flying more without ending the mission. Of course you'll
never top
the high score list without taking some chances.

Commonly Asked Questions

> Sometimes when I eject or crashland, there is a continuous tone that
> never fades. How do I turn it off?

Under certain conditions, the stall indicator sound can get stuck on. This DOS sound-driver problem does not apply to Redux.

> When first lifting off the ground, why does the nose lift for a second,
> and then seem to knock back down?

If you pull back too hard on the stick while lifting off, your tail wheel will be slammed back down onto the pavement, and when it bounces upward, your nose appears to bounce downward. It is possible to crash and burn as a result of this, so pull back on the stick gently.

> Why does the stall light go on sometimes when I'm taxiing on the runway?

The stall detector is looking at the airflow over the wings. It is calibrated to detect stalls when the plane is flying. When on the ground,
the weight of the plane is held by the landing gear, thus making the stall
detector give incorrect readings. Just ignore it.

> Why doesn't the plane steer worth a bean when I'm on the runway?

Don't try to use the joystick to steer a grounded plane. The control stick left/right controls mostly only the roll (left/right tilt) of the plane. You must use the rudder controls to steer on the ground. These are the 'x' and 'z' keys, or the '7' and '9' keys on the keypad.

> Sometimes I can't get back into my plane. I just go right through it.

There are two possible reasons for this. One is that you must walk at least about a couple plane lengths away before attempting to reenter. This hysteresis is to avoid popping back into your plane before you had a chance to walk away.

The DOS version could also lock planes after walking far away; that renderer-dependent restriction does not apply to Redux. Historically, If
you stray so far from your plane that it has shrunk to a point, then the plane will engage the automatic security system, and lock itself up tight. In none of the missions is it imperative to walk anywhere near this far from your plane.

> Why don't the brakes appear to work?

In Corncob, the brakes do NOT toggle on/off. To stop you must hold down the brake key until the plane comes to a complete stop. The default brake binding is Period or keypad 0.

> How does the T key accelerate time?

Hold T to run the whole simulation ten times faster, on foot or in

Commonly Asked Questions (continued)

flight. Enemies, projectiles and production advance at the same rate; they no longer sleep while you travel. Release T for normal speed.

> Why does using the rudder in flight just make me stall?

Unlike other flight sims, Corncob's flight model includes the capability of your plane sliding sideways through the air. The rudder causes your nose to flip to the side, without necessarily changing the direction of motion of the plane. Using too much rudder (by holding the key down too long), will cause the plane to turn to the point where it is hurling sideways through the air. Note that as soon as you release the rudder, the sidewind makes the nose tend to flip back into the sidewind, thus restoring the original orientation of the plane. The rudder is useful for flipping the nose a little to the side for a quick shot at a side target without the need of negotiating the slow process of turning the plane.

> How do I know when to end the mission?

A large part of Corncob is bringing your plane back to the base. Therefore you must bring your plane to a friendly airstrip, or else when you go to fly again, you will have one less plane in that theater.

To exit without losing a plane, choose End current sortie from the pause menu when you are flying low over your airfield (with the radar tower flashing white), or else land on the air strip. Landing on the airstrip will give you a much higher score multiplier, so it is worth an attempt if you think you can do it without crashing.

To find an airfield, follow the homing indicator that runs along the bottom of the windscreen.

> How do I zoom down closer when in map mode?

When in map mode, the KEYPAD 9 key will zoom down, and the KEYPAD 3 key will zoom up. Use the joystick or keypad arrow keys to move around. To make the map jump back to your present position, hit the KEYPAD 7 key.

> Why do I die every time I eject, regardless of altitude?

Ejecting successfully from a flying plane requires two keys. E to eject, and Q to pull your ripcord. If you don't pull your ripcord, you will die when you strike the ground.

Commonly Asked Questions (continued)

If you bail out at 20000 feet, wait until you get closer to the ground before pulling your ripcord, otherwise you will be stuck in the air for a long time.

> Why are some generator orbs seemingly indestructible?

Some orbs have been designed to be invincible to normal weapons, and the only effective way to take them out is via suiciding the plane into them. Make sure you bail out before contact, or else you will be killed in the ensuing explosion. If you must bail a good way away from the target, hit caps lock to control your plane remotely.

The indestructible orbs have bases which are a light shade of blue/green.

> Why does the complete light refuse to go on, even though I know
> the mission is completed?

The mission objectives have to be given some geometric scope via the mission builder when the mission is created. Almost all mission complete sensing is done via the generator orbs. IE, if you have completed a mission, but the complete light doesn't go on, make a flyby of the mission's generator orb, (or its remains) and many times the complete light will belatedly light up. Note that if your complete light never lights, you will not get credit for accomplishing an objective.

> Sometimes my plane goes into a weird mode where my keypad arrow
> keys stop working, and something weird happens.

Numlock toggles the keypad arrow keys between views, and control of the plane. Note that the numlock light on the keyboard will not change. If you would like to use the views without changing the mode of the arrow keys, hold down the left control key to temporarily change the function of the arrow keys to view, the normal function of the keys returns when you release the left control key.

> I can't find any way to put down the landing gear!

The plane has landing gear that comes down automatically. From a game standpoint, it seemed unnecessary to have something which doesn't add to the fun of the game, and can cause pilots to be killed. We all get punished in life enough for being absent-minded, why be punished when playing a

Commonly Asked Questions (continued)

computer
game?

> Why can this Corsair do a loop directly after take off?

Because the number one priority in this game was to have fun. That meant
increasing the thrust of the engine by a factor of four. For those sim buffs
who will criticize that, how many flight games keep the realistic figure of
10-15 minutes just to climb to 17,000 feet? Who wants to spend 1/4 hour of
their life just sitting there watching your computer plane climb?

CORNCOB OTHER WORLDS CAMPAIGN ORDER FORM

[Historical DOS documentation - retained for reference]

CORNCOB OTHER WORLDS CAMPAIGN ORDER FORM

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Long term game playability: Few Levels w/ hard difficulty Lotsa Levels

CORNCOB OTHER WORLDS CAMPAIGN ORDER FORM (continued)

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The Corncob program itself

Corncob 3D was written by Kevin Stokes, with a lot of additional programming by George Welch. The program documentation was written by Kevin Stokes. Original music was written by Jim Collymore, A.S.C.A.P. All music in the game is copyright 1986-1991 James L. Collymore, and the use of the three music files for any purposes except for Corncob 3D is strictly illegal. The music is contained in the following files: DESTINY3.ROL, RIDINGT.ROL, and DREAMS0.ROL. These files cannot be legally copied or distributed without the author's permission.

The VGA directory (created the first time Corncob 3D is run) contains all the digitized artwork in the game. These photos were extensively retouched using the excellent program MVP Paint, by David Johndrow. You can order your own copy of this extremely useful program from MVP Software.

Corncob 3D was written in assembler and C. The excellent library of routines Spontaneous Assembly was used in developing the program.

This software is based in part on the work of the Independent JPEG Group.

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